

Second International Workshop on Perception-driven Graphics and Displays for VR and AR (PerGraVAR)

Martin Weier*

RheinMain University of Applied Sciences

Oliver Staadt†

University of Rostock

Bipul Mohanto‡

University of Rostock

Colin Groth§

MPI Saarbrücken

Alexander Marquardt¶

Bonn-Rhein-Sieg University of Applied Sciences

Corentin Salaün||

MPI Saarbrücken

Praneeth Chakravarthula**

UNC Chapel Hill

ABSTRACT

The PerGraVAR workshop aims to foster a collaborative environment for researchers and developers to discuss perceptual findings, identify challenges, and present the latest research in the field of perception-driven rendering and computational displays. Thereby, we expect the participants to gain new ideas for research, establish potential collaborations with other researchers focusing on similar topics, and create a better understanding of best practices in setting up and performing user studies and experiments. The workshop covers a variety of topics that exploit and explore perceptual and cognitive limitations and potentials. Key areas include advancements in display technologies, interactive rendering techniques, and eye-tracking systems, as well as their optimization to align with human capabilities. This exchange of knowledge is essential for driving innovation in software and hardware systems.

Index Terms: Virtual Reality, Augmented Reality, Visual Perception, Computational Displays

1 INTRODUCTION

Over the past decade, high-quality Extended Reality (XR) systems have become widely available on a consumer level. While pixel densities and the dynamic range increase, the refresh rates are getting higher and higher, and display latencies are continually reduced. Currently, novel technologies such as displays with a growing number of displayed views per pixel (ranging from stereo, multi-view, holographic, lightfield displays, optical see-through, and video pass-through displays) have entered the market. These advancements not only enhance visual experiences but also highlight the need for improved interoperability across XR platforms, addressing challenges in ensuring consistent performance and user experience across diverse devices. Likewise, multi-layered displays and adaptable lenses are integrated into head-mounted devices. Emerging prototypes, such as retinal displays and bionic contact lenses, are paving the way for future innovation.

All these advances in display technologies will tighten the requirements for image synthesis techniques. Today, displaying realistic-looking graphics at high refresh rates is already computationally demanding, especially for high pixel densities, a wide Field of View (FoV), and when rendering in stereo. Also, as the size of datasets continually increases, a larger FoV makes it possible to present more of this information at once.

*e-mail: martin.weier@hs-rm.de

†e-mail: oliver.stadt@uni-rostock.de

‡e-mail: bipul.mohanto@uni-rostock.de

§e-mail: cgroth@mpi-inf.mpg.de

¶e-mail: alexander.marquardt@h-brs.de

||e-mail: csalaun@mpi-inf.mpg.de

**e-mail: cpk@cs.unc.edu

It is evident that this divergence between display technologies and rendering techniques will cause significant issues on the technical side (e.g., limited computing power, bandwidth, battery life, power consumption, and heat generation) and can have severe implications on the mental and cognitive load. Understanding these implications is crucial for developing systems that enhance user experience without overwhelming users. However, the visual system has some limitations and the highest possible visual quality is not always necessary. The knowledge of those limitations can be used to develop more efficient rendering systems, e.g., by foveating pixel density. Today, displays allow for more direct and complete addressing of the human visual system and eye-tracking hardware enables to perform active measurements of the gaze. Therefore, knowledge about perceptual processes is gaining increasing importance. The PerGraVAR workshop allows for discussing those techniques and systems that take advantage of the human visual system that are more tailored to the human visual system.

2 WORKSHOP OBJECTIVES

The primary objectives of the PerGraVAR workshop are to foster a collaborative environment for researchers, developers, and other stakeholders to exchange knowledge, share insights, and identify key challenges in the fields of perception-driven graphics and computational displays. To this end, the workshop offers a platform for discussing the following research topics:

- Innovative XR display devices and interoperability challenges in XR graphics
- Rendering and information visualization methods that target or exploit perceptual potentials and limitations, including gaze-contingent rendering and interaction techniques
- Studies that provide insights into perception & cognition models and processes
- Perception-driven image metrics and compression strategies
- Validation methodologies, benchmarks, and measurement methods, including eye-tracking, saliency, attention models Just noticeable differences, signal thresholds, and biases
- Novel measurement and processing techniques such as autorefractors or wavefront sensing
- Experimental designs and techniques for conducting user studies
- Systems, findings, and general issues related to specific perceptual issues such as:
 - Depth-of-Field
 - Vergence-accommodation conflict
 - Stereo disparity manipulation
 - Wide Field-of-View

3 WORKSHOP AGENDA

The workshop is structured as a half-day event. It will consist of the two keynote talks (1h30mins) and the paper presentations (1h30mins). Also, we plan on preparing a panel discussion. The workshop will focus on identifying pressing perceptual and cognitive challenges in XR and exploring hardware and software solutions to address these issues. We will also discuss prior work done on perception-driven graphics and perceptual display in VR and AR, drawing parallels, highlighting open questions, and exploring addressable potentials.

4 KEYNOTES

For the second part of the PerGraVAR workshop, two keynotes and a panel discussion will form the basis for further discussions. This year's workshop will feature keynote speeches by:

Ana Serrano is an Assistant Professor at Universidad de Zaragoza, Spain. Previously she was a Postdoctoral Research Fellow at the Max-Planck-Institute for Informatics working with Prof. Karol Myszkowski. She obtained her PhD under the supervision of Prof. Diego Gutierrez and Prof. Belen Masia at the Graphics & Imaging Lab (Universidad de Zaragoza). During her PhD she was the recipient of an Adobe Research Fellowship (honorable mention) in 2017, and a NVIDIA Graduate Fellowship in 2018. Her thesis was awarded with one of the Eurographics 2020 PhD awards. Recently, she received the Eurographics 2023 Young Researcher Award and the VGTC VR 2024 Significant New Researcher Award.

Qi Sun is an Assistant Professor at New York University. Before joining NYU, he was a research scientist at Adobe Research. He received his PhD at Stony Brook University. His research interests lie in VR/AR, perceptual computer graphics, computational display, and applied perception. He is a recipient of the IEEE Virtual Reality Best Dissertation Award. With colleagues, his research has been recognized as several best paper and honorable mention awards at ACM SIGGRAPH, IEEE ISMAR, IEEE VR, IEEE VIS, and ACM SAP.

5 ORGANIZERS

Martin Weier is a full professor for visual computing at the Hochschule RheinMain in Wiesbaden, Germany. He received his MSc in Computer Science from the Bonn-Rhein-Sieg University Sankt Augustin and his Ph.D. from Saarland University, Germany. His research interests include perception and gaze-contingent rendering where he published a series of papers that deal with perception-driven techniques and gaze-contingent ray tracing systems in head-mounted displays. Martin's other research interests include visual analytics, high-quality rendering approaches, compiler technologies, and virtual and augmented reality display systems.

Oliver Staadt is a full professor of Computer Science, Director of the Institute of Visual and Analytic Computing, and was, from 2016 to 2018, Dean of the Faculty of Computer Science and Electrical Engineering at the University of Rostock. He received a Master of Science in Computer Science and a PhD in Computer Science from TU Darmstadt and ETH Zurich, respectively. Prior to joining the University of Rostock, he was an Assistant Professor of computer science at the University of California, Davis, where he was also the director of the Virtual Reality Laboratory. His research interests include computer graphics, virtual and augmented reality, computer vision, telepresence, and visualization. He is a Fellow of Eurographics and a member of IEEE Computer Society and ACM SIGGRAPH.

Colin Groth received his MSc and Ph.D. in Computer Science from Technische Universität Braunschweig, Germany in 2020 and 2024 respectively. Currently, he is working as a Postdoc in the Computer Graphics group of Hans-Peter Seidel at Max Planck Institute. His research interests include perceptual-driven techniques in VR, particularly for mitigating cybersickness. With his methods and research, he also likes to encourage cross-disciplinary collaboration and develop applications for real-time rendering, image processing, and augmented reality.

Bipul Mohanto is a PhD student in the Visual Computing and Visual Analytics research group at the University of Rostock, Germany. He holds a Master's degree in Computer science from the University of Eastern Finland, Finland, and the University Jean Monnet, France (Erasmus Mundus). His research focuses on perception-driven real-time physically-based rendering and the applied human perception in VR.

Alexander Marquardt is a postdoctoral researcher at Bonn-Rhein-Sieg University of Applied Sciences (BRSU) in Germany. He received his MSc from BRSU and his PhD in Computer Science from the University of Bremen. His research focuses on developing multisensory user interfaces that enhance interaction and perception in XR environments, aiming to improve user experience and system effectiveness.

Corentin Salaün is a PhD student in Hans-Peter Seidel's computer graphics group at the Max Planck Institute in Germany. He holds a master's degree in computer science from the University of Rennes 1. His research focuses on perception-guided physically-based rendering for offline and real-time applications.

Praneeth Chakravarthula is an Assistant Professor of Computer Science at UNC Chapel Hill. His research interests lie at the intersection of optics, perception, graphics, optimization, and machine learning. Praneeth is a Senior Member of Optica, recipient of IEEE VR Best Dissertation Award, Timothy L. Quigg Inventor of the Year at UNC, and several Best Paper and Demo Awards. He completed his postdoctoral research at Princeton University, obtained PhD in computer science from UNC Chapel Hill, and B.Tech and M.Tech degrees in Electrical Engineering from IIT Madras.

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